

Enhancement Mode GaN FETs and ICs Visual Characterization Guide



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A detailed description of the EPC enhancement mode transistors and integrated circuits physical characteristics is given including the visual criteria all devices must meet before they are released for shipment to customers. This article, used in conjunction with the two companion articles, "Assembling eGaN FETs"¹, and "EPC GaN Transistor Parametric Characterization Guide"², gives the user a set of tools to develop circuits and systems that take advantage of the enhancement mode GaN FET's and IC's advanced form factor and consequent unprecedented performance potential.

OVERVIEW OF GALLIUM NITRIDE (GaN) TECHNOLOGY

In June of 2009, Efficient Power Conversion Corporation (EPC) introduced the first enhancement mode gallium nitride on silicon power transistors designed specifically as power MOSFET replacements. These products were developed to be produced in high volume at low cost using standard silicon manufacturing technology and facilities. For more information about EPC's GaN technology, go to www.epc-co.com.

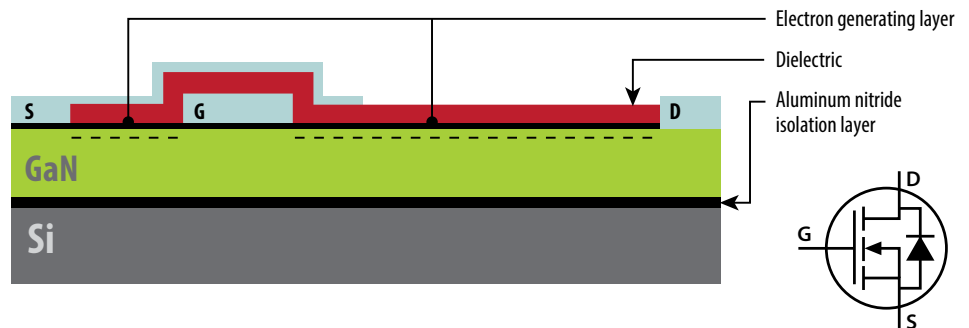


Figure 1: EPC's GaN transistor structure

STRUCTURE

A device's cost effectiveness starts with leveraging existing production infrastructure. EPC's process begins with silicon wafers. A thin layer of Aluminum Nitride (AlN) is grown on the Silicon to isolate the device structure from the substrate. On top of this AlN, a thick layer of highly resistive GaN is grown. This layer provides a foundation on which to build the active transistor. An electron generating material comprised of Aluminum, Gallium, and Nitrogen (AlGaIn) is applied on top of the GaN. This layer creates an abundance of free electrons just below it. Further processing forms a depletion region under the gate. To enhance the transistor, a positive voltage is applied to the gate in a similar manner to turning on an n-channel, enhancement mode power MOSFET. A cross section of this structure, repeated many times to form a

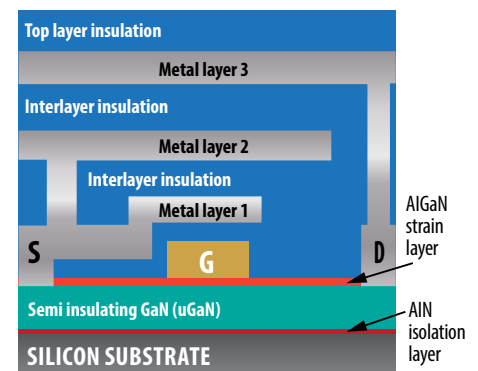


Figure 2: Device construction

complete power device, is depicted in figure 1. The end result is a fundamentally simple, cost effective solution for power switching³. EPC's GaN transistors are lateral devices with all three terminals: gate, drain, and source, on the top side of the chip. Generally, EPC devices have three layers of metal used to connect the active device to the outside world (fig 2). The top metal layer is then used as a foundation for solder bumps as shown in Figure 3. This configuration allows EPC's GaN transistors to eliminate unnecessary elements of traditional power MOSFET packaging that contribute to higher inductance, thermal and electrical resistance, higher costs, and compromised reliability.

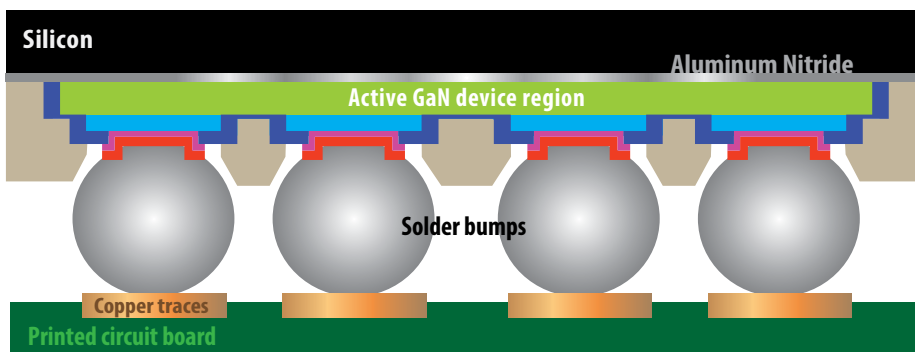
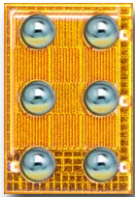


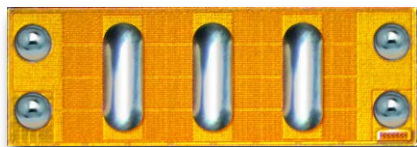
Figure 3: Flip chip

This section includes bump and side views of selected EPC's GaN transistors and integrated circuits. Solder bars or solder balls are used to make reliable connections directly to a printed circuit board. A polyimide coating on top of multiple silicon dioxide and silicon nitride layers are used to seal the active device from the outside environment.

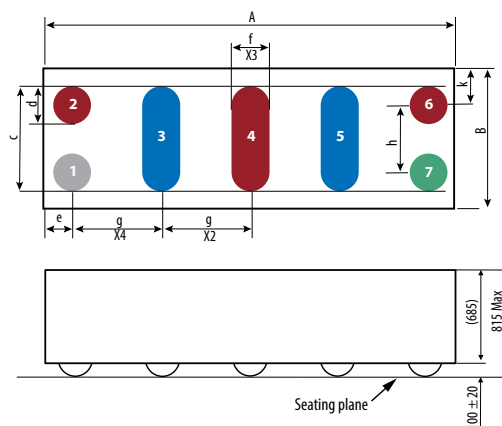
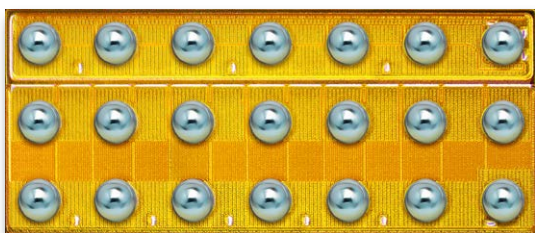
EPC2040



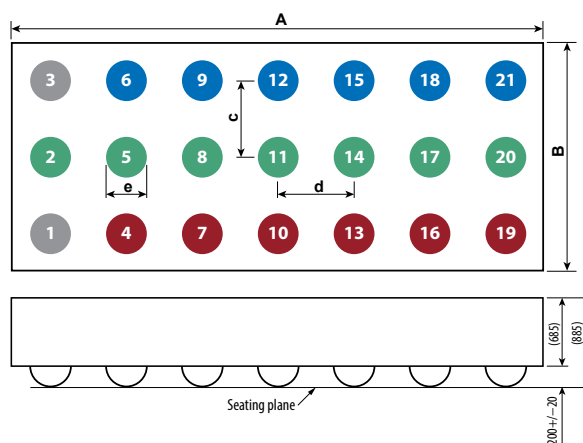
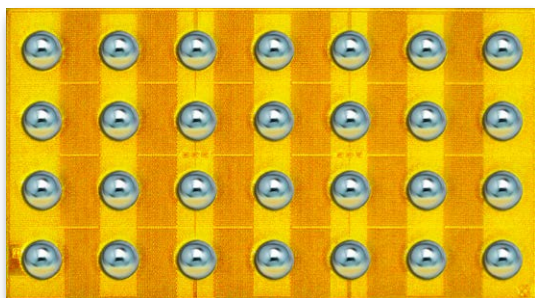
Technical drawing of a 6-seater aircraft cabin cross-section. The diagram shows a rectangular cabin with width A and height B. Six seats are arranged in a single row. Seats 1 and 2 are grey, 3 and 4 are blue, and 5 and 6 are red. Dimensions include seat width f, aisle width e, seat pitch g, and cabin width X2. A side view shows the seating plane and a height dimension of 815 Max.

EPC2019

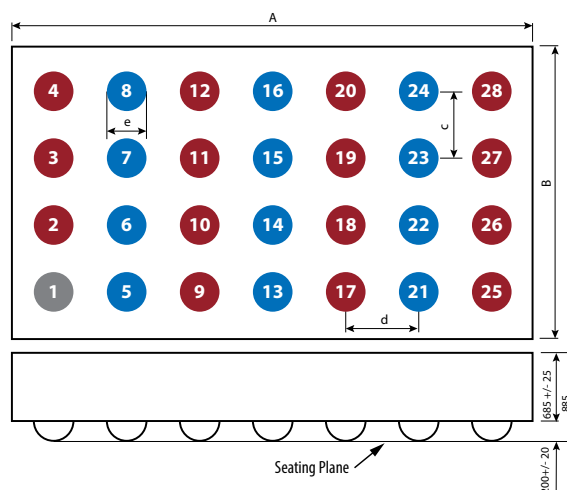
2.77 mm x 0.95 mm
 Max size (μm):
 2796 x 980

**EPC2111**

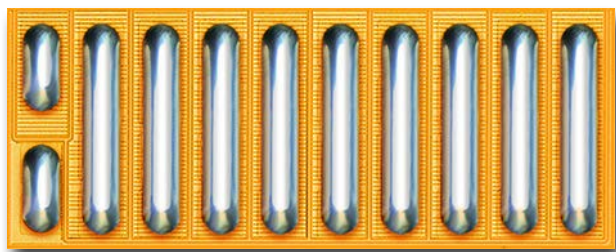
3.5 mm x 1.5 mm
 Max size (μm):
 3530 x 1530

**EPC2053**

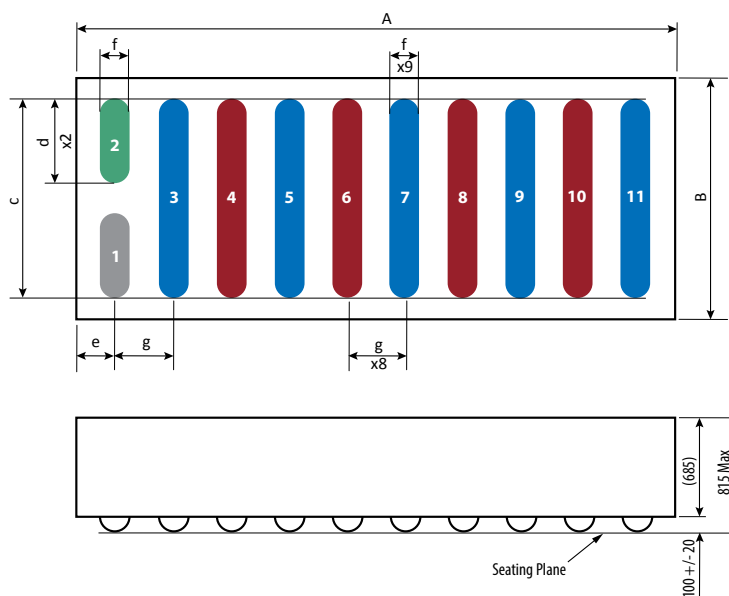
3.5 mm x 2 mm
 Max size (μm):
 3530 x 1980



EPC2015C EPC2001C

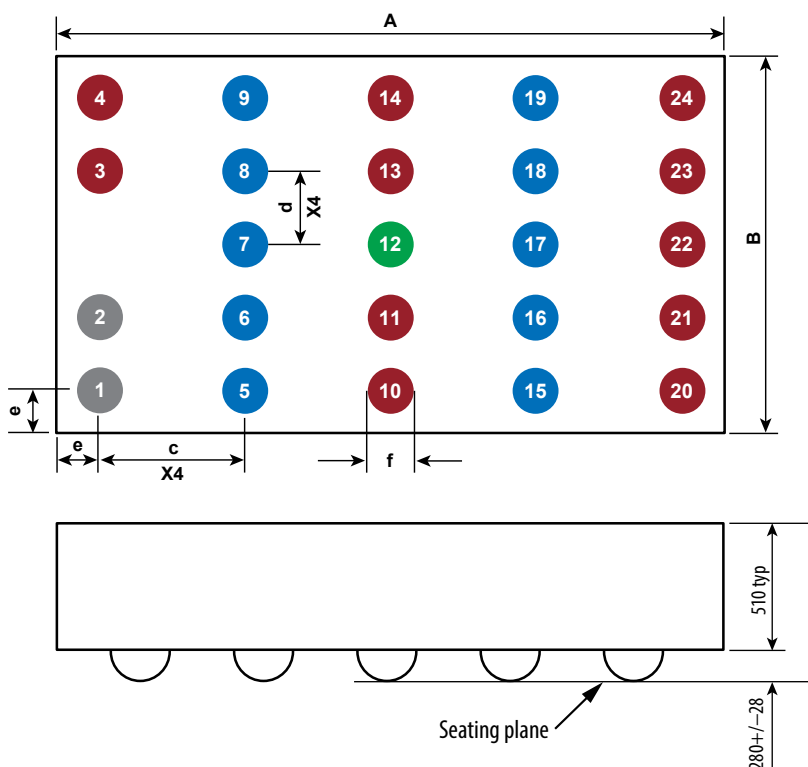
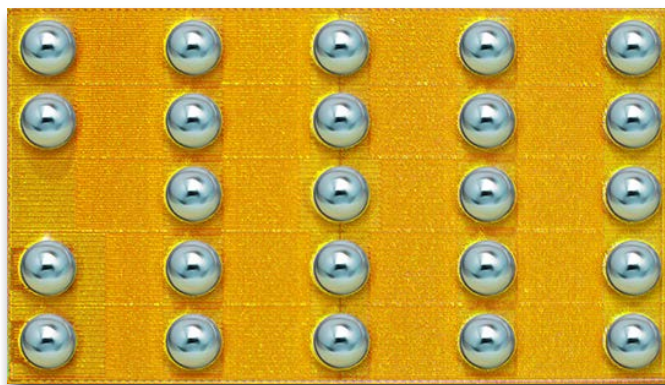


4.1 mm x 1.6 mm
Max size (μm):
4135 x 1662



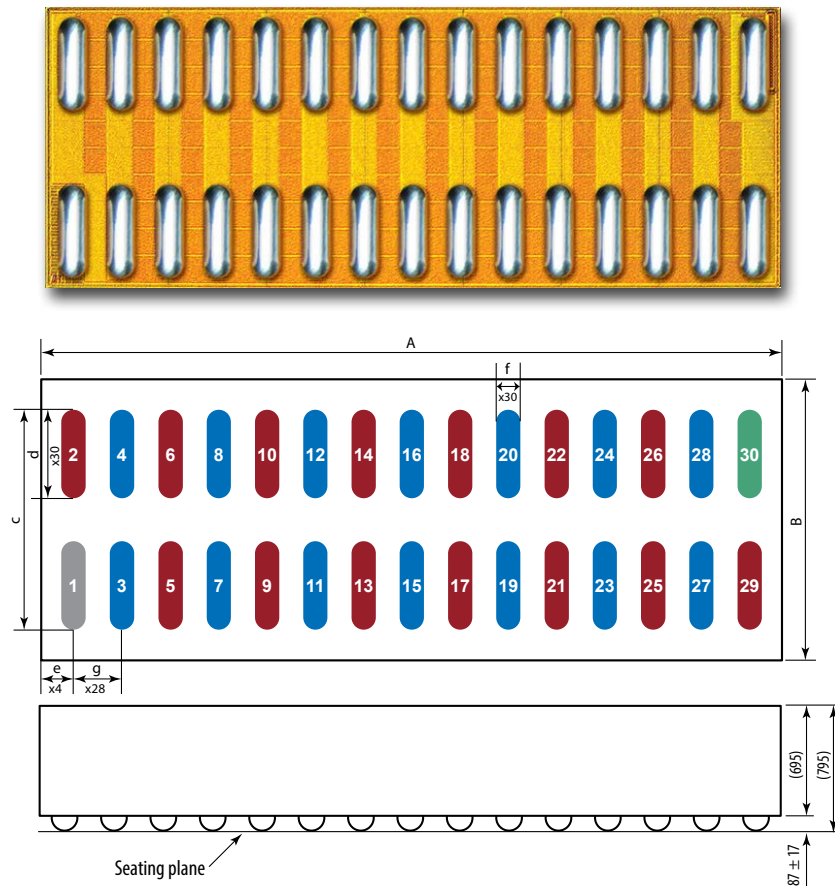
EPC2029 EPC2030 EPC2031 EPC2032 EPC2033 EPC2034 EPC2034C

4.6 mm x 2.6 mm
Max size (μm):
4135 x 1662



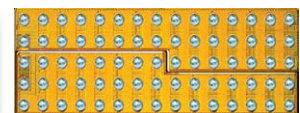
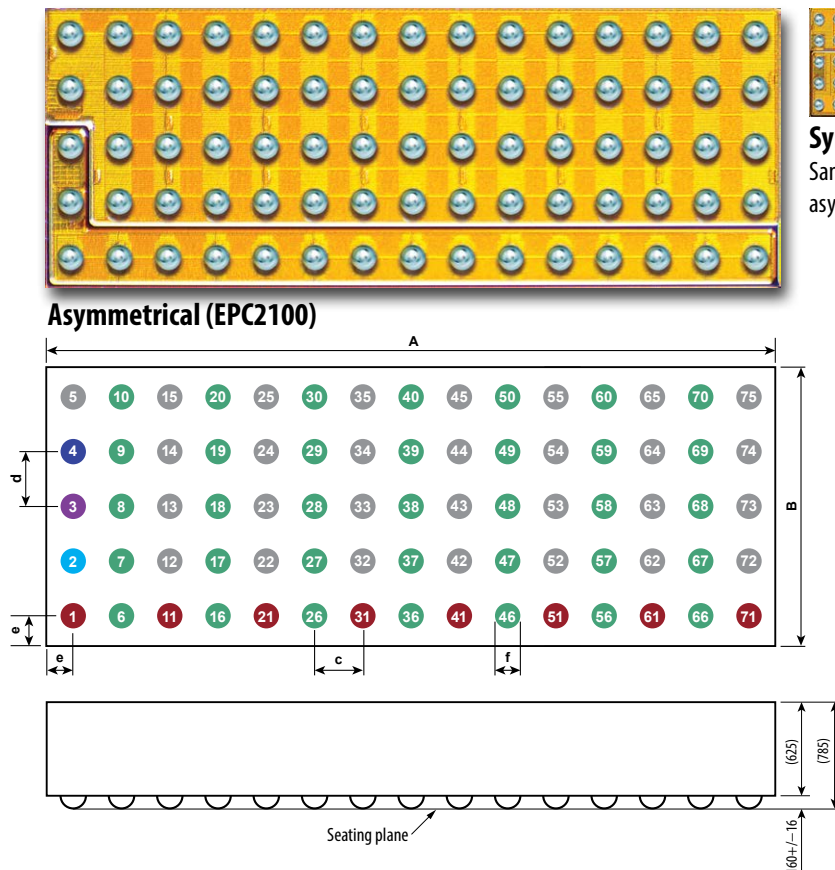
EPC2020
EPC2021
EPC2022
EPC2023
EPC2024
EPC2206

6.05 mm x 1.6 mm
Max size (μm):
6080 x 2330



EPC2100–Asymmetrical
EPC2101–Asymmetrical
EPC2102–Symmetrical
EPC2103–Symmetrical
EPC2104–Symmetrical
EPC2105–Asymmetrical

6.05 mm x 2.3 mm
Max size (μm):
6080 x 2330



Symmetrical
Same die size as the
asymmetrical configuration

This section showed a selection of EPC's eGaN FETs and ICs. For a full listing of products please see the website at: epc-co.com/epc/Products/eGaNfetsandICs

EPC designed these devices such that they are quite robust. In figure 4 are some examples of devices that were intentionally tested after extreme mechanical abuse.

Figure 4a, b, and c: EPC transistors with extreme mechanical damage successfully completed long term reliability testing. It should be noted that the substrate is not electrically active with respect to the device.



Fig 4c: EPC1001 part (above) passed 1000 Cycle TC -40°C to 125°C. Die width ~1900 μm
Back side chip >250 μm

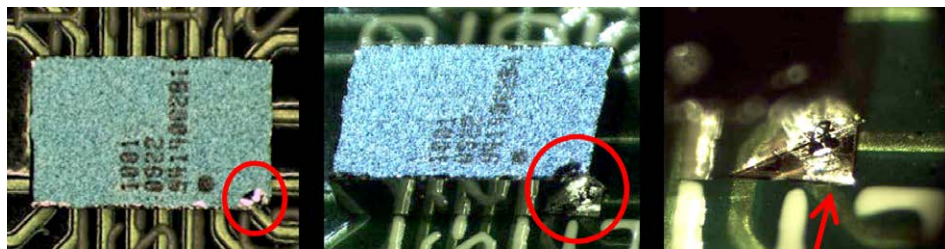


Fig 4a: Visual reject part shown in pictures above – EPC1001 part above passed 1000 hours HTRB 100 V bias 125°C. Back side chip extends down to the front of the chip. **Far right:** (area indicated with arrow) Unsupported chip structure hanging out over corner passed 1000 hours HTRB.



Fig 4b: Visual reject part shown in pictures above – EPC1001 part with severe corner chip. Back side chip extends all the way through the chip. Die was still functioning in application. Die is shown post dismount from PCB board.

The edge protection structure reduces the risk of damage to the active portion of the device. Figure 5 is a set of microscope images of the typical edge structures and, for clarity, the original design plots are also shown to the left. Care should be taken during assembly such that chips cannot touch the polyimide edge seal ring.

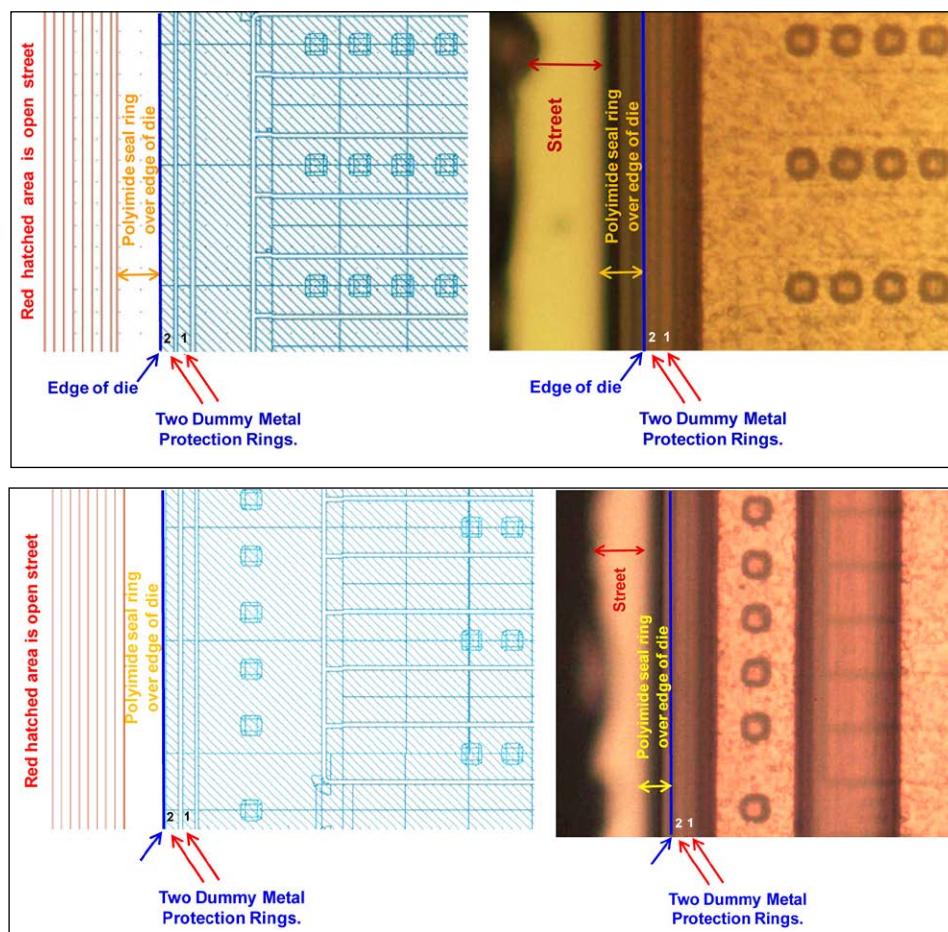


Fig 5: EPC's die top surface is designed to reduce the probability that chips will intrude into the active portion of the device. Pictured is the edge structure which includes two dummy metal rings and a polyimide coating.

SHIPMENT CRITERIA

Prior to shipment, 100% of EPC devices are electrically tested and visually inspected. Tables 1a and 1b show a summary of the visual Acceptance/Rejection criteria used as of the date of this writing. Tables 2a and 2b, over the following pages, contain several examples of both good and bad die measured against these criteria.

Table 1a: Front side visual inspection criteria

Die Front Side Visual Inspection Category	Rejection Criteria
Amount of remaining street	Amount of remaining street can be zero providing no chip is beyond second (outer) metal dummy ring
Chips	Chip touches seal ring
Damaged Solder Bumps	Solder drag or displacement >50% of original distance between bumps; defect which reduces bump height by 30% of original height
Foreign material	Foreign material linking any two bumps
Contamination /defect / metal residue / bridging material (can appear as stains or severe discolorations)	Any contamination, defect, metal residue or bridging material that is >50% of original distance between bumps or eliminates the separation between adjacent metal lines
Ink dots	Ink on front of chip
Missing Bumps	Any missing bumps
Particles	Reject if particle reduces the original distance between bumps by 50%
Polyimide peeling, incomplete	Area bridging two adjacent top metal layers
Probe Marks	Reject if no probe marks
Probe Needle Drag	Solder drag or displacement >50% of original distance between bumps or crossing metal lines
Scratches	Scratches through the polyimide > 25% of the width of the metallization block under the bump

Table 1b: Back Side Visual Criteria

Note : substrate is not electrically active with respect to the device

Die Back Side Visual Inspection Category	Rejection Criteria
Chips	Nominal chip thickness without the bump is listed in datasheet for each device. MIL883 says reject if chip is >50% of total die thickness deep. Post datecode 1944* the reject chip extends a distance/depth of 100 µm or more away from the edge of the die.
Correct back side marking for part number EPCXXX, first line of back side laser mark will be XXXX	Reject if EPC part number is not correct in first scribe line or if marking is not legible
Orientation in pocket	Incorrectly oriented in pocket (locator dot should be on side near carrier tape holes)
Particles	Specification in development
Scratches	Specification in development
Stains / Discoloration / Burn Marks	Laser mark not readable

*Note: Prior to datecode 1944, reject if chip is > 250 µm wide

TABLE 2a: EPC die front side visual examples of Accept/Reject categories

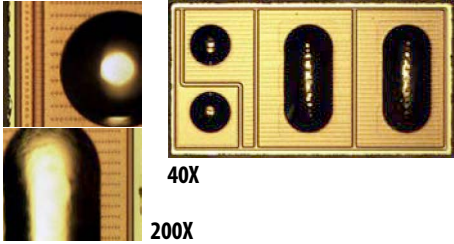
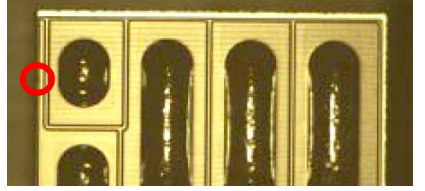
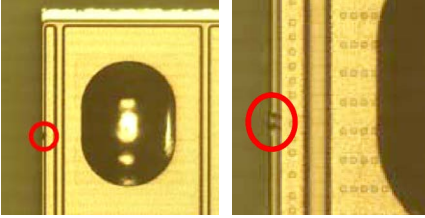
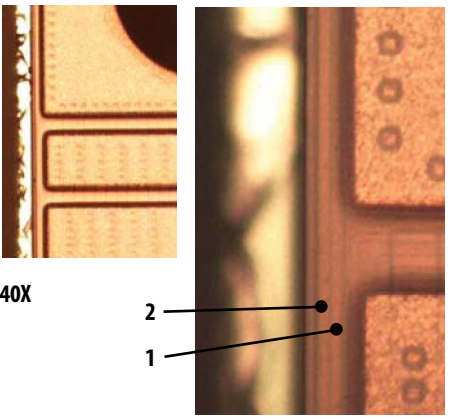
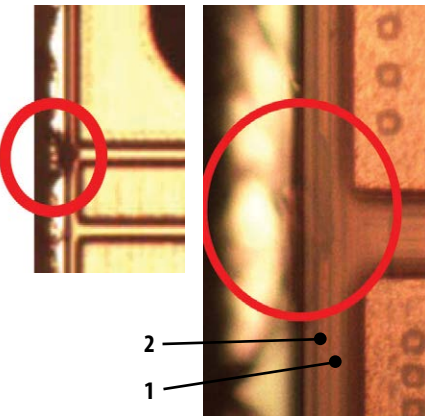
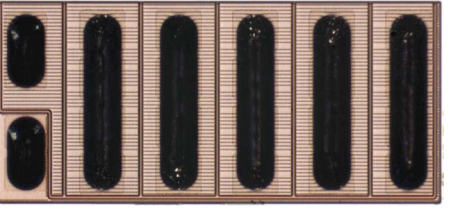
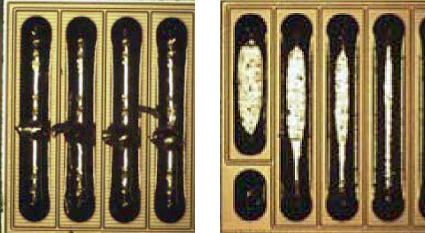
Amount of remaining street Reject Criteria: Amount of remaining street can be zero providing no chip is beyond second (outer) metal dummy ring. <i>See section on chips</i>	ACCEPT  40X 200X Street (thin white line) seen all the way around die in above pictures. Remaining street width can vary.  40X 200X No street visible at 40X on left side of die, but no chips into the outer metal dummy ring at 200X	REJECT  No street visible at 40X on left side of die. Chip can be seen. This chip extends past both metal dummy rings when checked at higher magnifications  100X 200X Chip on left side can be seen at 100X is past both dummy metal rings. At 200X, defect is shown certainly past both the dummy metal rings.
Chips Reject Criteria: Chip cannot touch edge seal	ACCEPT  40X 2 1 At 400X, showing the two metal dummy rings in one die edge design type	REJECT  2 1 Shown at 400X. Defect is past the outer the metal dummy ring #2.
Damaged solder bumps Reject Criteria: solder drag or displacement >50% of original distance between bumps; defect which reduces bump height by 30% of original height	ACCEPT  Clean die	REJECT  Solder drag Squashed bumps

TABLE 2a: EPC die front side visual examples of Accept/Reject categories (*continued*)

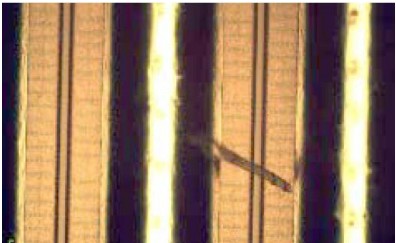
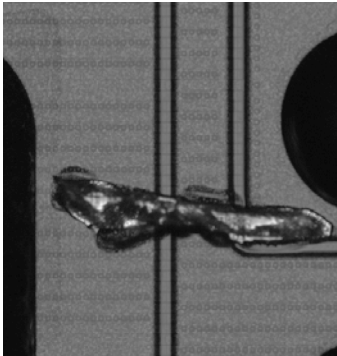
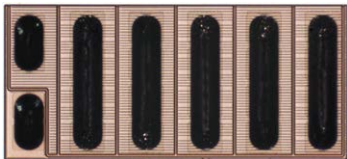
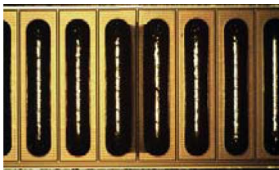
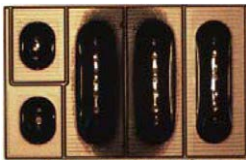
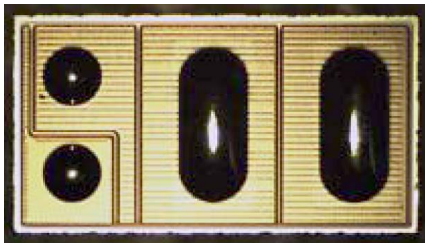
Foreign material	ACCEPT	REJECT
Reject Criteria: Foreign material linking any two bumps	 No foreign material (street not shown in picture)	 Foreign material bridging bumps
Contamination /defect / metal residue / bridging material	ACCEPT	REJECT
Reject Criteria: Any contamination, defect, metal residue or bridging material that is >50% of original distance between bumps or eliminates the separation between adjacent metal lines (can appear as stains or severe discolorations)	 No bridging, no contamination (street not shown in picture)	 [Picture courtesy of KYEC]
Ink dots	ACCEPT	REJECT
Reject Criteria: Ink on front of chip	  No ink on these chips	  Ink dot seen on chips
Probe marks	ACCEPT	REJECT
Reject Criteria: Reject if no probe marks	 This die has probe marks on the bumps	 No probe marks – this is a reject

TABLE 2a: EPC die front side visual examples of Accept/Reject categories (*continued*)

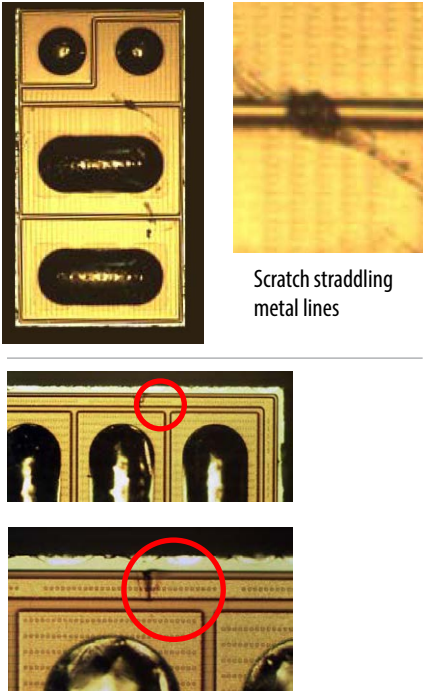
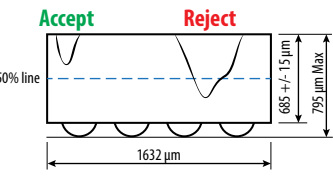
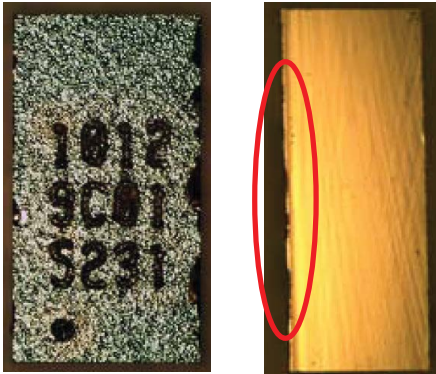
	ACCEPT	REJECT
Scratches Reject Criteria: Scratches through the polyimide > 25% of the width of the metallization block under the bump or bridging metal lines	 No scratches	 Scratch straddling metal lines Scratch cuts across all the metal dummy rings and first metal 3 ring

TABLE 2b: EPC die back side visual examples of acceptable and rejectable categories

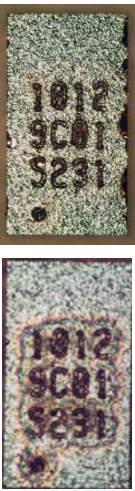
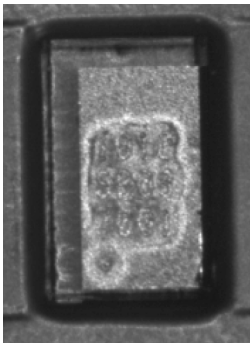
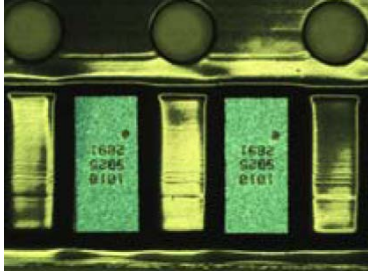
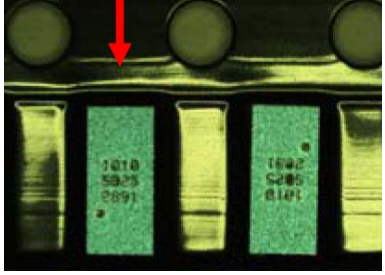
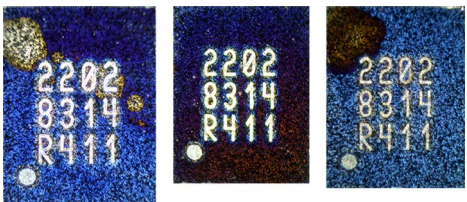
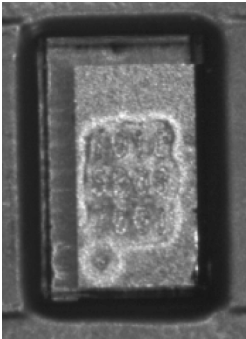
Note : Substrate is not electrically active with respect to the device

	ACCEPT	REJECT
Back Side Chips Reject Criteria: Nominal chip thickness is listed in the datasheet for each device. MIL883 says reject if chip is >50% of total chip thickness deep. Post datecode 1944*, the reject chip extends a distance/depth of 100 μm or more away from the edge of the die. Die Back Side Chipping Details Nominal chip thickness is listed in the datasheet for each device. MIL883 says reject if chip is >50% of total die thickness deep. Post datecode 1944*, the reject chip extends a distance/depth of 100 μm or more away from the edge of the die. Example against part number EPC2007C that has max length of 1632 μm is shown below: 	 Die is > 0.9 mm wide Chips < 100 μm wide Same die as at left: Backside chips are very shallow	 Die is 1.6 mm wide, chip is >100 μm wide. Lower left corner is broken down to front side of die

*Note: Prior to datecode 1944, reject if chip is > 250 μm wide

TABLE 2b: EPC die back side visual examples of Accept/Reject categories (continued)

Note : substrate is not electrically active with respect to the device

	ACCEPT	REJECT
Correct back side marking for part number EPCXXXX, first line of back side laser mark will be XXXX* Reject Criteria: Reject if EPC part number is not correct in first scribe line or if marking is not legible The second and third lines of the back side laser mark are for lot traceability <i>*see datasheet for part marking spec of specific device</i>	 EPC1010: Top line is 1010  EPC1012: Top line is 1012 Color variation, but readable	 Unreadable [Pictures courtesy of KYEC]
Orientation in pocket Reject Criteria: Incorrectly oriented in pocket (locator dot should be on side near carrier tape holes)	 Carrier tape holes and locator dot are both on this side of tape	 Die on left is incorrectly oriented. Locator dot is on opposite side of carrier tape holes
Stains / Discoloration / Burn marks Reject Criteria: Laser mark not readable	 Laser mark readable  Color variation, but readable  Cosmetic backside discoloration	 Unreadable [Pictures courtesy of KYEC]

References:

- [1] epc-co.com/epc/documents/product-training/Appnote_GaNAssembly.pdf (AN009 – Assembling eGaN FETs and Integrated Circuits)
- [2] epc-co.com/epc/documents/product-training/Characterization_guide.pdf (AN004 – Enhancement-Mode GaN Transistors)
- [3] epc-co.com/epc/documents/product-training/Appnote_GaNfundamentals.pdf (AN002 – Fundamentals of Gallium Nitride Power Transistors)